

**Knowledge Graphs and Ontologies: Research Services 2nd
Call for Proposals**
Closing date: 11 August 2026

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Summary

[The Alan Turing Institute](#)'s Defence & National Security Grand Challenge is seeking to commission research projects that will contribute towards meeting UK Government agencies strategic requirements for systematic sensemaking at scale and pace.

The projects will need to be delivered between the period 01 October 2026 – 31 March 2027.

There may be an opportunity for the project to be extended into a second phase (to take place between April 2027 – March 2028). Please provide details of an optional extension as part of your application. Please note funding for optional extensions is not guaranteed and any optional extension will need to be agreed, confirmed in writing with associated agreements updated. A decision on whether the project will be extended into the second phase will be made at the end of phase 1.

Background

Knowledge graphs and ontologies are becoming increasingly important tools in the analysis of information and production of knowledge in many national security applications. The volume, variety, velocity, and veracity of available data, and the information it yields, has reached a point where systems are required to fuse information from many different types of data, support, and provide timely accurate analysis. Defence and intelligence require the tracking of a fast-evolving world and must contend with information variety and veracity at scale. Whilst there have been significant advances in linked data and knowledge graph technologies, multiple challenges remain to being able to deploy systematic sensemaking at scale.

The Requirement

We are seeking to commission 3-6 small-scale research activities that are initially supported as pilot studies. Those that show promise may subsequently be further developed into full studies with the parties concerned. Pilot studies will likely involve 2-3 researchers and have a duration of 3-6 months. Studies that extend into the second phase may involve more participants and may extend over 1-3 years.

A Turing Principal Researcher from the Defence and Security team will work closely with the commissioned academic groups to ensure close collaboration between the UK Government research teams and those in the academic community.

We would expect research activities to be related to, but not necessarily restricted to one or more of the following areas of interest:

- Robust and efficient event and relationship extraction. Comprehensive population of knowledge graphs from text across multiple languages will require significant improvements in the standard, performance and efficiency of event and relationship information extraction. This in turn may require more consistent annotation and new datasets for training extraction models.
- Complex (multi-part) entity modelling. Entities of interest to national security are typically complex, comprising many parts, many of which may be replaced. Many things will only be of interest only when they act as a particular part of a larger entity;

the latter being the subject of interest, but the former is observed. Efficient methods for representing and inferring part-whole relationships both intra and inter multi-part entities with temporal sensitivity would therefore be particularly useful capabilities.

- Event co-reference and linking. As with entity resolution there is a need for linking different references to events to the event of interest, the complication being that events typically comprise sub-events be part of wider events. Events are inherently complex entities, requiring care over how parts relate in bounding and linking what is considered an event of interest.
- Knowledge Graph Datasets and Evaluation. As knowledge base applications grow in range, size, and complexity of application, so will the need for comprehensive evaluation of proposed techniques, solutions, and systems. This in turn requires the development of appropriate test and evaluation datasets and frameworks. Datasets geared towards complex events and entities, and scenarios where competing narratives give rise to inconsistent knowledge (competing hypotheses) are particularly sought.
- Argumentation, provenance and decision-making formalisation and integration. KGs in national security applications are expected to include not only information from observation sources, but from analysis, inference, and assessments. Sources may have questionable veracity. Representation of provenance for sources, assessment, decision making, and the argumentation applied in KS will therefore be important.
- KG inference (reasoning-based link and node prediction). The ambition of systematic sensemaking is for machines to infer knowledge about the world, including what might be (with assessment of confidence). We will need methods to combine symbolic reasoning with other KG analytics to provide and propose additions to knowledge graphs and knowledge bases.
- Reasoning with uncertainty. It follows from the need to semantically represent uncertainty surrounding information in a KG, that reasoning takes it into account. Systematic reasoners operate on the logic of factual entailment. We will need methods and tools to provide reasoning in the presence of uncertainty to give reliable qualification for inference and decision making.
- KG aware HCI. Visualisation and interaction with knowledge graphs as graphs is known to be counter-productive above a few hundred nodes. KG applications are typically concerned with information within the graph rather than the graph as a whole. We will need different HCI modalities for consumers of KGs, such as multiple connected views, multi-perspective summarisations, and agentic AI integration. Techniques, and evaluations thereof, for effective human exploration, discovery, and analysis of knowledge with contextual awareness are significant requirements for sensemaking at scale.
- Scale. Formal knowledge graphs are inherently complex. Performant techniques will need to be developed and evaluated for all stages in systematic knowledge graph construction, persistence, maintenance, and analysis. This is true not only for areas of interest outlined above, but for all supporting functions.

Also of interest:

- Entity resolution and linking. There are many entities, and many different identification schemes and identifiers used to distinguish them. Accuracy, and specificity, of data varies. Ensuring information is correctly associated requires the entities concerned to be resolved. Solutions will need to address the scale, range, and uncertainty inherent in open and multi-source KG applications.
- Adaptive (ephemeral) entity modelling (user driven ontology extensions). Schemas and domain ontologies define classes with use-cases in mind. However, to be truly flexible, models should be able to create class definitions to meet a user's analytic requirement at the time it arises.

- Formal representation of uncertainty: Uncertainty with respect to information arises from various phenomena in the production thereof. Systematic KG analysis will need to factor in uncertainty, which in turn suggests grounding in an ontology that includes uncertainty modelling. Exploration, development, and establishment of uncertainty analysis frameworks, building upon recent foundational work, will be needed.
- IES and Ontology standards maturation. UK Government agencies have invested in early research collaborations to develop models to address some of the challenges of tracking uncertain information represented across multiple schemas with varying quality to derive high precision knowledge leveraging well-founded domain ontologies. The Information Exchange Standard (IES) is emerging from this research and proving useful in early implementations. However, it, as with specific domain ontologies, will need further maturation and integration with the wider semantic technology eco-system for its contributions to be sustainably supportable in industrial scale applications. We would encourage research activities to be in the defence and/or national security context.
- Domain Ontology development. Although various domain ontologies exist, some of which are relevant to areas of concern for defence and security, further work is required to enhance and enable interoperability and their application in analytic application. Alignment with or extension of the IES would be encouraged as a route for cross-domain interoperability.
- Ontology grounding. Data and information exist in many forms, both unstructured and structured. Much will not conform, or ever will conform, to a single ontology. Methods for high accuracy systematic ontological grounding and translation would address a significant bottleneck for information fusion with knowledge graphs. As with domain ontologies, use of IES as a target for grounding would be welcomed, but methodology is objective of interest rather than chosen framework.
- Skills market development. Knowledge graphs and ontologies are highly specialised areas, and skills are correspondingly rare. There is a need to grow the pool of skilled knowledge technology specialists and engineers to support the growth and adoption of formal knowledge-based applications.

We would encourage research activities to be in the defence and/or national security context. Applications from other fields are welcome, providing they can provide a rationale for why/how approach might translate into defence and/or security applications.

Please note, all resources (datasets, models etc.) that a proposal would look to use must be available for government use.

Available Funding

Funding between £40k-70k (excluding VAT), depending on scope, will be available for a limited number of projects over a duration of 5-6 months, between October 2026 to March 2027. The research will be funded at Full Economic Cost and VAT will apply. The proposal should be defined such that the project can be completed by 31 March 2027.

Eligible costs include:

- Salary of personnel working directly on the project – this could include, for example, PIs, postdoctoral research associates, research assistants, data managers, data scientists or software engineers.
- Overheads, estates and indirects.

- Travel and subsistence for project researchers (e.g., attending conferences, travelling to/from the Turing/other collaborators).
- Conference or event attendance fees (where conference/event is directly applicable to the research project).
- Cloud computing or other high-performance computing costs.
- Other costs which are specifically justified for the project e.g., books, meeting room or catering costs, specific laptops.
- Open access publications.

Requested costs must be pre-approved by your organisation, so please ensure to include this statement of approval as part of the [Submission Approval Letter](#).

Please note, the budget requested may not be funded to the maximum requested amount. The Turing may work with applicants to build collaborative projects which may involve updating costs for projects.

Terms and conditions

The funding will be made available under The Alan Turing Institute's Defence & National Security programme Research Service Agreement terms and conditions. For a copy of the terms please contact Agne Clemente by emailing aclemente@turing.ac.uk.

You will be required to confirm your organisation's acceptance of these terms as part of this application process.

Please note:

- All Foreground IPR and any jointly Generated IPR shall vest in Turing in order for Turing to fulfil its obligations to UK Government Agencies.
- All research works for public dissemination (e.g., research articles, conference papers) require approval from both UK Government Agencies and Turing prior to submission. The UK Government agencies have a vested interest in the award holder publishing their research. Accordingly, requests will not be unreasonably refused.

Eligibility

To be eligible to apply you must:

- Be part of a UK university or research institute. Commercial organisations that are in partnership with Academia may apply but should take note of our IP terms. Overseas universities or institutes are not eligible.
- Have permission from your organisation to apply, i.e., ensure your organisation can agree to the Terms and Conditions provided and that you include as part of the application an approval of submission letter from your research/contracts/finance office stating this. An example of a letter is provided.
- We expect applicants to seek the relevant ethics approvals from their institute prior to starting the project.
- There will be national security checks for shortlisted applicants. Successful applicants
 - will need to meet UK Governments baseline personnel security standard (see Government baseline personnel security standard - GOV.UK before contracts can be awarded. Please provide full names, dates of birth and nationalities for all researchers you plan to submit for your research proposal (if a researcher

holds multiple nationalities, please list all of them). personnel security standard (see [Government baseline personnel security standard - GOV.UK](https://gov.uk/government/publications/government-baseline-personnel-security-standard)).

How to apply

Applications must be submitted via the online portal at <https://ati.flexigrant.com/>. If you have not already done so, all applicants must first register on the system and provide basic details to create a profile. If you have any questions regarding the application form or using the online system, please contact the Research Project Manager, Agne Clemente, aclemente@turing.ac.uk.

Please use the budget template provided in the Flexigrant application form. Your approval of submission letter should also confirm that your research/finance office have reviewed that the costs provided are correct.

The submission approval letter must confirm that:

- if not already covering the entire period of the project, then the contract of employment for project researchers will be amended and/or extended as necessary to enable the successful completion of the project.
- that the researcher is already employed by the university and no recruitment is needed to fulfil this project.
- the project will be given full access to the facilities, equipment and personnel as required by the application.
- the costs included in the application have been correctly calculated with the support of the Research Office / Finance Department (or equivalent).
- the terms and conditions of the agreement have been reviewed by the Research Office / Legal Department (or equivalent).
- the letter signatory is authorised to approve the submission of applications for funding and the application has met all internal approval procedures.

The Principal Investigator must ensure the same is received for all collaborators / universities on multi-party applications.

We must receive your application by **1600 hrs 11 August 2026**.

What should be in the proposal?

The proposal should:

- Be in OpenDocument, MS Word or PDF
- Describe the scope and technical approach of the proposed work –
- This should be a narrative description of the principals/solutions the project would aim to achieve and how those solutions may relate to the requirement section.
- Describe how the approach would lead to the desired results (and any real-world impact).
- Include a description of how the task is decomposed, thematically i.e. by work package, include optional extension at the end.
- For each work package, what activities will be undertaken to produce the results.
- Answer questions such as:
- What is innovative about the proposed research?

- Why are you uniquely placed to undertake this task?
- What is the expected scientific impact?
- How do you intend to demonstrate this impact?
- Identify any key risks and mitigations.
- Reference related work or/and relevant experience.
- Include FEC cost.

If you are employed by a University that is a member of the [Turing University Network](#), please contact your Turing Liaison ([a list of Turing Liaisons is available on the Turing website](#)) to make them aware of your application. They can provide support, answer questions and involve you as part of the Turing community at your university from now on.

Deliverables

Deliverables will be confirmed prior to contracts being signed; however they should include:

Ref.	Deliverable	Due by	Format	Description
<i>D1</i>	Project Kick off Meeting	31/10/2026	Meeting	Meeting to initiate the project
<i>D2a,b</i>	Progress update meeting	15/12/2026 14/02/2027	Meeting	Progress update meeting with funders
<i>D3</i>	Report	31/03/2027	Report	Report covering the pilot study, approach, methods, results etc
<i>D4</i>	Presentation of Pilot Study	31/03/2027	Presentation of framework / tool	Final presentation of the research
<i>D5</i>	Project Closure Meeting	31/03/2027	Meeting	Meeting to close the project and agree follow on steps

In addition to the above, other research outcomes from commissioned tasks will be expected to be delivered as specified for individual projects. This may include media, and/or data resources. It is also expected that research outcomes from commissioned tasks that are to be published are done so through reputable open and publicly available media where appropriate, as per standard academic practice, and after approval from Turing and funding partners as explained above.

Assessment and review

Following eligibility checks, proposals will be reviewed by an assessment panel who will rank the proposals based on score.

The assessment panel will consider the following criteria:

- Quality: This will consider the method and concepts for the proposed research. This will assess if the methods are suitable for delivering the desired outputs and pushing forward fundamental understanding in the field.
- Viability: This will assess how feasible it is to practically carry out the proposed research, and if it can be delivered in the time frame. This will account for the difficulty

of the tasks, logistical factors surrounding delivery, and the track record of the proposed research team.

- Significance: This will consider the relevance to the call and the themes that are represented. Consider whether the proposed research is likely to deliver real-world and/or research impact
- Justification of resources: This will consider whether the proposal is appropriately resourced and suitable expenditure has been included in the budget.
- Knowledge and expertise: Candidates should demonstrate they fulfil the bullet points in the requirement section.

Each of the criteria will be scored and while all criteria will have equal weighting in evaluation, there will be a minimum requirement on significance to be considered for approval.

Key Dates

Deadlines are as follows

Activity	Date
Call open	29 June 2026
Proposals to be Submitted*	11 August 2026*
Announcement of Results	04 September 2026
Earliest Research Project Starts by**	01 October 2026**
Research Completed and deliverables submitted by	31 March 2027

*Proposals must be submitted via Flexigrant by **1600 on 11 August 2026**.

**Any project agreements not signed 15 working days prior to deadline start date above may result in funding offer being withdrawn and going to an application on the reserve list.

Post-award information

Government furnished assets

A baseline view on narrative shall be shared, in a workshop with researchers at or near the beginning of agreed work; noting that we expect the definition to evolve with researcher input.

Project meetings

Successful applicants will be expected to attend a kick-off meeting and a project close meeting, with a Technical Partner from the Defence & National Security programme Partner/s. Applicants will also be expected to join regular project progress meetings. These may take place online, at the Turing, at UK Government partner site, or at the project lead's university.

Publications

Please note, approval from the Defence & National Security team is required prior to publication; in such cases, approval will not be unreasonably withheld.

Reporting and dissemination

Extracts from reports may be collated into update papers for the D&NS Programme Board, Strategic Partners Board, Turing Innovations Ltd Board, and the Turing's Trustee Board.

Awardees may also be required to present their work to members of the D&NS team, the D&NS Programme Board and/or other invited audience during the award period.

Reporting allows further identification and signposting of potential additional opportunities for the benefit of the awardees and the Turing; for example, opportunities from across the Turing's network such as new collaborations, external/public engagement, media/press, other funding availability, speaking slots at or invitations to events/conferences/seminars.

Queries

Please contact Agne Clemente, The Alan Turing Institute, Research Project Manager at aclemente@turing.ac.uk.